

Solving Systems Algebraically by Substitution or Elimination

Solving by substitution, eliminating by adding or subtracting, and eliminating after scaling one or both equations

Before you solve, decide how.

- **Substitution** is fastest when one equation already gives a variable by itself, or gives it after one easy step.
 - **Elimination by adding or subtracting** works when a variable has the same coefficient, or opposite coefficients, in both equations.
 - **Elimination after scaling** is the general method: multiply one or both equations so a variable's coefficients match, then add or subtract.
 - Every problem here names two real quantities. Write what x and y stand for, *with their units*, before you write either equation, and check your pair in **both** original equations at the end.
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1. School play. A school play sold 40 tickets in all. The number of child tickets sold was 3 times the number of adult tickets sold.

Let x be the number of adult tickets and y the number of child tickets, so

$$x + y = 40, \quad y = 3x.$$

Solve by substitution. How many of each kind of ticket were sold?

Answer: _____

2. Garden. A rectangular garden has a perimeter of 36 metres. Its length is 4 metres more than its width.

Let x be the length in metres and y the width in metres, so

$$2x + 2y = 36, \quad x - y = 4.$$

Rearrange the second equation and solve by substitution. Give the length and the width in metres.

Answer: _____

3. Boat and current. A boat travels downstream at 18 km/h and upstream at 12 km/h. Downstream the current helps the boat; upstream it holds the boat back.

Let x be the boat's speed in still water and y the current's speed, both in km/h, so

$$x + y = 18, \quad x - y = 12.$$

The y terms are already opposites. Add the equations to eliminate y , then find both speeds.

Answer: _____

4. Cafe order. Four muffins and three juices cost 17 dollars. Four muffins and five juices cost 23 dollars.

Let x be the price of one muffin and y the price of one juice, in dollars, so

$$4x + 3y = 17, \quad 4x + 5y = 23.$$

Nothing needs multiplying here. Explain in one line which operation — adding or subtracting — removes x , then carry it out and find both prices.

Answer: _____

5. Stationery. Three pens and two notebooks cost 16 dollars. Five pens and four notebooks cost 30 dollars.

Let x be the price of a pen and y the price of a notebook, in dollars, so

$$3x + 2y = 16, \quad 5x + 4y = 30.$$

Multiplying *one* equation is enough to line up a variable. Say which equation you scale and by what, then solve.

Answer: _____

6. Ticket sales. A concert sold 200 tickets in total. Adult tickets cost 5 dollars each and student tickets cost 3 dollars each, and the sale brought in 780 dollars.

Let x be the number of adult tickets and y the number of student tickets, so

$$x + y = 200, \quad 5x + 3y = 780.$$

Scale the first equation so that one variable cancels, then solve. How many tickets of each kind were sold?

Answer: _____

7. Two gyms. Gym A charges a 20 dollar joining fee plus 5 dollars a month. Gym B charges a 50 dollar joining fee plus 2 dollars a month.

Let x be the number of months and y the total cost in dollars, so

$$y = 20 + 5x, \quad y = 50 + 2x.$$

Both equations already give y , so substitution is one step: set the two expressions for y equal. After how many months is the total cost the same, and what is that cost?

Answer: _____

8. Crates. Four crates of apples and three crates of pears weigh 30 kilograms. Three crates of apples and five crates of pears weigh 28 kilograms.

Let x be the weight of an apple crate and y the weight of a pear crate, in kilograms, so

$$4x + 3y = 30, \quad 3x + 5y = 28.$$

Here no coefficient matches, so *both* equations must be scaled. Choose a variable, write the two multipliers you will use, and solve.

Answer: _____

9. Two receipts. One receipt says two sandwiches and three drinks cost 19 dollars. A second receipt from the same shop, on the same day, says four sandwiches and six drinks cost 34 dollars.

Let x be the price of a sandwich and y the price of a drink, in dollars, so

$$2x + 3y = 19, \quad 4x + 6y = 34.$$

Try to eliminate a variable. Write down exactly what happens, say what the result tells you about the two receipts, and explain what it means for these two lines that they never meet.

Answer: _____

10. Challenge — sports equipment. A school buys 3 footballs and 2 basketballs for 31 dollars, and 5 footballs and 4 basketballs for 55 dollars.

Let x be the price of a football and y the price of a basketball, in dollars.

(a) Write the system and solve it by elimination. Give both prices, and check them in both original purchases.

(b) A supplier's catalogue claims that 6 footballs and 4 basketballs cost 60 dollars. Show that this claim cannot be true alongside the two purchases above, and explain what it means for three equations in two unknowns to have no solution that fits all three.

Answer: _____